Paper 10 — Why Neural-Network Sharpness Changes During Training

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Universal Sharpness Dynamics in Neural Network Training: Fixed Point Analysis, Edge of Stability, and Route to Chaos

Gradient dynamics | Fixed-point analysis | Edge of stability and chaos

Key: `kalra2025universal` | **Tags:** D, G, A / R, B | **Tier:** Must read (main paper)

The paper in one minute

Question: Why does neural-network sharpness sometimes fall, later rise, and eventually oscillate near the edge of stability? Can it become chaotic?
Model/Method: Study discrete gradient descent on a two-layer *linear* UV network with one training example. Rewrite its updates using a residual and a curvature-related variable, then analyze fixed points and their stability.
Finding: Initial conditions and parameterization determine whether training shows early sharpness reduction, progressive sharpening, and edge-of-stability oscillations. Above a model-specific learning-rate threshold, an oscillatory manifold can become attracting; increasing the learning rate can produce period doubling and chaos.
Why we care: Sharpness is a *dynamical variable*, not a fixed property of the loss landscape. Large-step training can oscillate without immediately diverging, but the toy model's dynamics do not transfer unchanged to real networks.

1. Start with the four training regimes

The paper organizes training into four stages. *Not every network exhibits all four*: initialization scale, learning rate, and parameterization change which stages appear.

Stage	What happens to Hessian sharpness $\lambda_{\max}(H)$?
T1: early transient	Can decrease; at large step sizes loss and sharpness may temporarily spike (a "catapult").
T2: saturation	Sharpness changes little for a while.
T3: progressive sharpening	Sharpness grows as the network moves through parameter space.
T4: edge of stability (EoS)	Sharpness fluctuates near the classical quadratic boundary $2/\eta$.

2. The first mathematical intuition: why $2/\eta$ appears

Take the simple local quadratic $L(z) = \frac{1}{2}az^2$ with positive curvature a. Gradient descent makes

$$z_{t+1} = z_t - \eta a z_t = (1 - \eta a) z_t .$$

For the coordinate to contract, $|1 - \eta a| < 1$, which means $0 < \eta a < 2$. Exactly at $\eta a = 2$, the multiplier is -1: the coordinate flips sign on every step without contracting in the ideal quadratic. Above 2, its magnitude grows.

Important precision

This $2/\eta$ boundary follows from a *fixed quadratic approximation*. Neural networks are nonquadratic; their curvature changes during the update. Oscillation at EoS does *not* by itself mean the full neural-network training run has diverged.

3. The UV network: simple input-output mapping, nonlinear weight dynamics

The authors use a two-layer linear network with input $x \in \mathbb{R}^d$ and one target y:

$$f(x; U, v) = \frac{v^\top U x}{\sqrt{n_{\text{eff}}}}, \qquad n_{\text{eff}} = n^{1-p}, \quad p \in [0, 1], \qquad \mathcal{L} = \tfrac{1}{2}(f - y)^2.$$

Here U and v are both trainable. The input-output function is linear in x, but the product $v^\top U$ makes GD nonlinear in the separate weights. The parameter p interpolates between parameterization regimes; n_{eff} is the *effective width*.

4. Worked example: curvature changes even when the model is linear

Use one scalar input $x = 1$, one unit ($n_{\text{eff}} = 1$), and let $f(u, v) = uv$ with $y = 1$. Then

$$\mathcal{L}(u, v) = \tfrac{1}{2}(uv - 1)^2, \qquad \frac{\partial \mathcal{L}}{\partial u} = (uv - 1)v, \qquad \frac{\partial \mathcal{L}}{\partial v} = (uv - 1)u.$$

Step 1. Take $u = v = 0.10$ with $\eta = 0.2$. The prediction is $uv = 0.01$ and residual $r = -0.99$. Both updates are

$$u^+ = v^+ = 0.10 - 0.2(-0.99)(0.10) = 0.1198.$$

Step 2. Here the scalar tangent-kernel quantity is $\lambda = u^2 + v^2$; it rises from 0.0200 to $2(0.1198)^2 \approx 0.02870$. The curvature proxy *increases* even as the loss falls.

Step 3. Contrast $u = v = 2$ with $\eta = 0.1$. Now $r = 3$ and $u^+ = v^+ = 2 - 0.1(3)(2) = 1.4$. The same quantity decreases from $\lambda = 8$ to $\lambda^+ = 3.92$: *reduction of the curvature proxy*. Thus its direction of change depends on *where training starts*.

What the toy example does — and does not — show

It illustrates why a product-parameterized linear network can sharpen or flatten. It is *not* a proof that all realistic networks follow those trajectories, nor is $u^2 + v^2$ always equal to the top Hessian eigenvalue. In the paper the authors analyze $\lambda = \operatorname{Tr} H$ as a *proxy* for measured sharpness.

5. Reduce an enormous parameter problem to two variables

Let the residual be $r_t = f_t - y$ and let

$$\lambda_t = \frac{\|Ux\|^2 + \|v\|^2 \|x\|^2}{n_{\text{eff}}}.$$

The full UV dynamics can be represented by the following *closed* update equations in (r_t, λ_t):

$$r_{t+1} = r_t \left(1 - \eta \lambda_t + \frac{\eta^2 \|x\|^2}{n_{\text{eff}}} r_t (r_t + y)\right),$$
$$\lambda_{t+1} = \lambda_t + \frac{\eta \|x\|^2}{n_{\text{eff}}} r_t^2 \left(\eta \lambda_t - 4 \frac{r_t + y}{r_t}\right).$$

These are the paper's key technical reduction (Equations 1–2). By studying where both updates vanish, the authors identify fixed points and a zero-loss fixed line. They use the stability of these sets to explain when curvature falls, rises, or oscillates.

6. Why a saddle can cause reduction, then sharpening

In the two-variable diagram, the point $(r, \lambda) = (-y, 0)$ corresponds to all UV parameters being zero. It is a saddle of the gradient dynamics. Some initializations move *toward* that region: weight norms and the curvature proxy decrease. As training later leaves the saddle neighborhood, the curvature proxy starts to increase. The resulting pattern is

$$\boxed{\text{approach saddle} \longrightarrow \text{sharpness reduction} \longrightarrow \text{plateau} \longrightarrow \text{progressive sharpening.}}$$

This is one mechanism, not a universal claim that every loss plateau is caused by a saddle.

7. A critical step size for the edge of stability

The zero-loss fixed line ($r = 0$) has stable points where $\eta\lambda < 2$. The paper derives a *necessary* condition for its UV model to exhibit EoS:

$$\boxed{\eta > \eta_c, \qquad \eta_c = \frac{\sqrt{n_{\text{eff}}}}{\|x\| y}} \qquad (y > 0).$$

Worked numerical check. With $n_{\text{eff}} = 16$, $\|x\| = 1$, and $y = 2$, the threshold is $\eta_c = 4/2 = 2$.

- At $\eta = 1$, this necessary condition is *not* met, so the proposed UV EoS mechanism is excluded.
- At $\eta = 2.5$, the condition is met, but *EoS is not guaranteed*: training must still remain in a trainable regime and encounter the corresponding attracting dynamics.

For $\eta > \eta_c$, the paper analyzes a line segment in the (r, λ) plane that acts as an *EoS attractor*. Rather than settling at a fixed point on the zero-loss line, the discrete trajectory can keep oscillating nearby.

Do not confuse a necessary condition with a sufficient guarantee

The paper also notes a maximum trainable learning rate $\eta_{\max}$ dependent on initialization. If $\eta_{\max} < \eta_c$, then EoS cannot be observed before divergence.

8. Where the "route to chaos" comes from

On the EoS manifold the model reduces further to a *one-dimensional cubic map* in the residual. Increasing η changes the map's attracting trajectory: one repeated value can become a two-cycle, then four values, then more complicated irregular fluctuations. This is called *period doubling*.

For an intuition-only analogue, consider $F(z) = \frac{1}{4}(z^2 - 1)^2$ and perform GD:

$$z_{t+1} = z_t - \eta F'(z_t) = z_t + \eta z_t (1 - z_t^2).$$

The map is cubic. Around the solution $z^* = 1$, its local multiplier is $1 - 2\eta$; at $\eta = 1$ it reaches -1, the threshold where a fixed point can lose stability through a flip bifurcation. *This illustrative cubic is not the UV paper's exact Equation 3.*

The paper's *actual* EoS-manifold map is

$$r_{t+1} = r_t + \frac{\eta r_t}{\eta_c y}\left(\frac{\eta r_t}{\eta_c y} - 2\right)(r_t + y).$$

The bifurcation diagram in its Figure 3 shows the period-doubling route in the UV setting as learning rate rises.

9. What is established, and what is observed?

Evidence type	Takeaway
Toy-model analysis	Closed two-variable UV dynamics; classification of fixed points; critical EoS condition and EoS-manifold cubic dynamics.
Neural-network observations	FCNs/CNNs/ResNets display parameterization-dependent sharpness trends; Figure 4 shows an observed association between EoS and test accuracy in the investigated setting.
Synthetic versus real data	Clear period doubling appears for certain synthetic data; real-image data produce broad, correlated bands of sharpness fluctuations rather than the same clean bifurcation pattern (Figure 5).

10. Contributions, assumptions, limitations

Main contributions

- Unifies early sharpness reduction, intermediate saturation, progressive sharpening, and EoS in one analytically tractable model.
- Explains how fixed-point locations, stability, initialization, and parameterization change the training trajectory.
- Derives an EoS threshold in the UV model and relates its oscillations to a cubic-map period-doubling route.
- Tests selected model predictions in larger architectures on real and synthetic datasets.

Important assumptions Exact analytic dynamics use a two-layer *linear* network, one training example, squared loss, and *full-batch, discrete* GD. The main critical threshold assumes $y > 0$; initialization and effective width influence which dynamics are accessible.

Limitations The UV model does not explain nonmonotone *decrease* of loss at EoS in realistic networks, does not reproduce long-range correlations of real-data sharpness fluctuations, and can miss a loss catapult at small effective width. The clean route to chaos is not established for arbitrary nonlinear networks on real datasets. This is not a universal convergence theorem or a prescription that chaotic training improves generalization.

11. Survey relevance and reading priority

IV — Dynamics: This is a useful recent mechanistic treatment of step size, initialization, EoS, and nonmonotone trajectories. **III — Geometry:** It links Hessian statistics and saddle neighborhoods to dynamical motion. **VI / theory versus practice:** It sharply illustrates the gap between a controlled toy model and real architectures.

Read first: Sections 3–5 and Figures 1–3 (especially Equations 1–3, Result 1, and Figure 3). **Then skim:** Section 6 and Figures 4–5. **Refer to appendices:** only for the details of the threshold, EoS attractor, and experiments.

12. Quick reference and Papers Master note

Term	Meaning
Sharpness	$\lambda_{\max}(H)$, the largest eigenvalue of the loss Hessian.
UV curvature proxy	$\lambda = \operatorname{Tr} H$, analyzed as a tractable proxy; not identical to the top eigenvalue in general.
EoS boundary	$\eta\lambda_{\max}(H) \approx 2$ (classical positive-curvature quadratic stability threshold).
UV EoS condition	$\eta > \sqrt{n_{\text{eff}}}/(\|x\|y)$; *necessary*, not sufficient.
Period doubling	Fixed behavior changes to 2-cycle, 4-cycle, etc. as the learning rate increases.

Papers Master — short note

The UV model explains sharpness reduction, progressive sharpening, and EoS through fixed-point stability and parameterization-dependent trajectories. It derives a necessary EoS learning-rate threshold and exhibits period doubling/chaos in a controlled setting; realistic networks show only partially matching, data-dependent behavior.

Original paper: Kalra, He & Barkeshli, ICLR 2025. Mathematical examples in this guide are pedagogical adaptations; model results and limitations are checked against the authors' paper.